

The MovieLens dataset is a dataset of 100,000 movie ratings, from which the provided movielens dataset is sampled. In the data, the 1,000 users give ratings from 1 to 5 to movies chosen from a list of 1,700 titles. In the data matrix (denoted X), if user i rated movie j , then $X[i, j]$ is that rating (an integer from 1 to 5). If that user did not rate that movie, then $X[i, j] = 0$. Thus, most entries in the dataset are 0, because most users only rate a small number of movies. You may assume that every user has rated at least one movie with each of the five ratings; that is, in every row, there is at least one 1, one 2, one 3, one 4, and one 5.

Suppose we wanted to use a K Nearest Neighbor model to recommend movies to users using this data. That is, for a given user who has rated some movies, we want to recommend a movie we think they will rate highly, but have not yet seen.

a. What would be a good distance measure for this data? Why?

Cosine distance; it does not consider when the values are zero

b. How does the distance measure you chose handle the fact that most entries in the data matrix are 0?

The Cosine Similarity measure is based on the dot product between two vectors. Since missing values are represented as 0 in the data matrix, they do not contribute to the dot product and hence do not affect the similarity score.